

CHAPTER 6

Content Framework



This chapter is part of the GREEN and GOLD reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

The Content Framework is best thought of as a categorization framework. The primary purpose of the Content Framework is to structure the work products and models that make up the architecture and to describe the architecture products used in an organization. It promotes consistency and coherence in the deliverables produced by following the phases of the Architecture Development Method. The Content Framework – in its extended form – also provides a comprehensive checklist of architecture deliverables that can be produced and reduces the risk of gaps in the final architecture deliverables. In other words, the Content Framework helps an organization use standard architecture concepts, terms, and deliverables.

6.1. A Categorization Framework

The Architecture Development Method provides a way to develop and manage architectures within an organization. It consists of eight main phases and is preceded by a Preliminary Phase. Central to the Architecture Development Method is Requirements Management – not a separate phase, but a mechanism for continuously testing the architecture against

it. Each phase of the Architecture Development Method – with the exception of Requirements Management – describes a set of architecture work products.

Because not every organization is the same, it is strongly recommended that the TOGAF Standard be tailored to the requirements and needs of the organization so that the framework is a good fit for the type of organization and its way of working. The TOGAF Standard should be tailored to represent the organization in which it is used.

One of the first steps in establishing the architecture function is to determine the use of the Architecture Capability. Chapter 9 provides further details on this process.

An essential task when establishing the organization-specific Architecture Capability during the Preliminary Phase of the Architecture Development Method is to define the following:

- **An Understanding of the Types of Entities:** It is necessary to understand the types of entities used in the organization and the relationships between them. This is critical to the creation of the architecture description. The *Enterprise Metamodel* represents this information in the form of a formal model.
- **A Categorization Framework:** In order to structure the architecture descriptions, the work products used to express an architecture, and the collection of models that describe the architecture, a categorization framework needs to be used – in other words, the *Content Framework*.
- **The Specific Artifacts:** After both the metamodel with the entities to be used and the categorization framework have been created, the artifacts to be developed can be determined.

The Content Framework and the Enterprise Metamodel are closely related. Where the Enterprise Metamodel is a detailed representation of the entities that exist in the organization, the Content Framework is a kind of high-level summarization. A more fitting term is a *categorization framework*.

The content of the Content Framework consists of a collection of models that represent the architecture. The categorization mechanism of the Content Framework may be used to structure a representation of the Enterprise Metamodel, which, as was mentioned earlier, is a detailed representation of the entities used in an organization. The Content Framework is intended to [11]:

- **Describe a detailed model** of the architecture products used in an organization. Since not all organizations are alike, the models of the architecture products may differ per organization. This is why it is so important to tailor the architecture framework to the needs and requirements of the organization in which it will be used.
- **Promote consistency and coherence** in the deliverables produced by following the phases of the Architecture Development Method. Using the Content Framework ensures that every time a particular deliverable is needed, it is developed in the same way. The entities behind the Content Framework's categorization mechanism ensure that the same entities are always used and that those entities always mean the same thing.

- **Provide a comprehensive checklist** of architecture deliverables that can be produced. The Content Framework allows a checklist to be created for each category of entities, specifying which architecture deliverables can (or should) be produced.
- **Reduce the risk of gaps** in the final architecture deliverables. Because it has been determined in advance which models use which entities, the chance of gaps is minimal, if not non-existent. The condition for this to work is that careful thought is given to filling out the Content Framework before it is implemented.
- **Help an organization use standards** such as architecture concepts, terms, and deliverables. The predefined entities within the Content Framework categories ensure that there is no ambiguity in the use of the entities. Each of the entities is given a definition, the terms to be used, and the deployment of the entity within the architecture deliverables.

The Content Framework is a well-thought-out categorization mechanism that seamlessly connects to the phases of the Architecture Development Method. Partly because of this connection, the Content Framework is a very valuable addition to the TOGAF Standard and therefore to any organization.

6.2. Advantages of the Content Framework

One of the advantages of the Content Framework is that it is central to defining and organizing architectural work products throughout the architecture development process. It is designed to provide a more modern, flexible, and outcome-driven approach to Enterprise Architecture.

The Content Framework defines a categorization framework to be used to structure the architecture description, the work product used to express an architecture, and the collection of models that describe the architecture [11].

The Content Framework aims to be more adaptable to various organizational contexts while ensuring that all relevant architecture elements are covered comprehensively. It contains the following key aspects:

- **Modular Structure:** The TOGAF Standard is more modular than previous versions, allowing organizations to adopt only the parts of the framework that are relevant to their needs. This modularity also extends to the Content Framework.
- **Core Content Model:** The Content Framework provides a comprehensive content model (Figure 6-1) that defines the types of building blocks and artifacts that an architecture effort should produce. It includes core concepts such as actors, processes, applications, data entities, and technology components. This model is designed to ensure consistency across various architectural views (Business, Data, Application, and Technology).
- **Extended Content Model:** The framework also allows for extended content models that provide additional depth in specific areas, such as security, governance, or data management. This makes it easier for architects to tailor the framework to their organizational context.

- **Architecture Artifacts:** The Content Framework defines key artifacts like catalogs, matrices, and diagrams that help document different aspects of the architecture, for example, a Business Process Catalog or an Application Portfolio Matrix. These artifacts are designed to align with the phases of the Architecture Development Method.
- **Views and Viewpoints:** The framework emphasizes the creation of views and viewpoints, which are ways of looking at the architecture to address the concerns of different stakeholders. Viewpoints help define the conventions for constructing and using the views, ensuring consistency and relevance.
- **Alignment with Agile and Digital Transformation:** The TOGAF Standard has placed a stronger emphasis on supporting agile methodologies and digital transformation efforts. The Content Framework is designed to be used in conjunction with modern practices like DevOps and cloud architecture.
- **Guidance for Tailoring:** The TOGAF Standard includes extensive guidance on how to tailor the metamodel and artifacts to fit specific organizational needs. This makes it easier to apply TOGAF in diverse environments. Tailoring the TOGAF Standard is done by selecting, adapting, and using those parts of the framework that are valuable to the situation or organization. It establishes a set of tools and a common vocabulary.

- **Outcome-Focused:** A major shift in the TOGAF Standard when compared with earlier versions is its focus on outcomes rather than just processes. The Content Framework reflects this by emphasizing the creation of valuable architecture artifacts that deliver business outcomes, rather than just following a rigid structure.

The Content Framework differs from previous versions in *flexibility*, *integration*, and its emphasis on *outcomes*. The modular and adaptable nature of the TOGAF Standard makes the Content Framework more flexible than in previous versions. It better integrates with other frameworks and methodologies such as the agile methodology. There is a stronger focus on creating actionable outputs that directly contribute to achieving business objectives.

6.3. Mapping to the Architecture Development Method

Looking at the Content Framework from a high-level perspective, it is not that difficult to see the implementation of the Architecture Development Method. Figure 6-1 illustrates the phases of the Architecture Development Method.

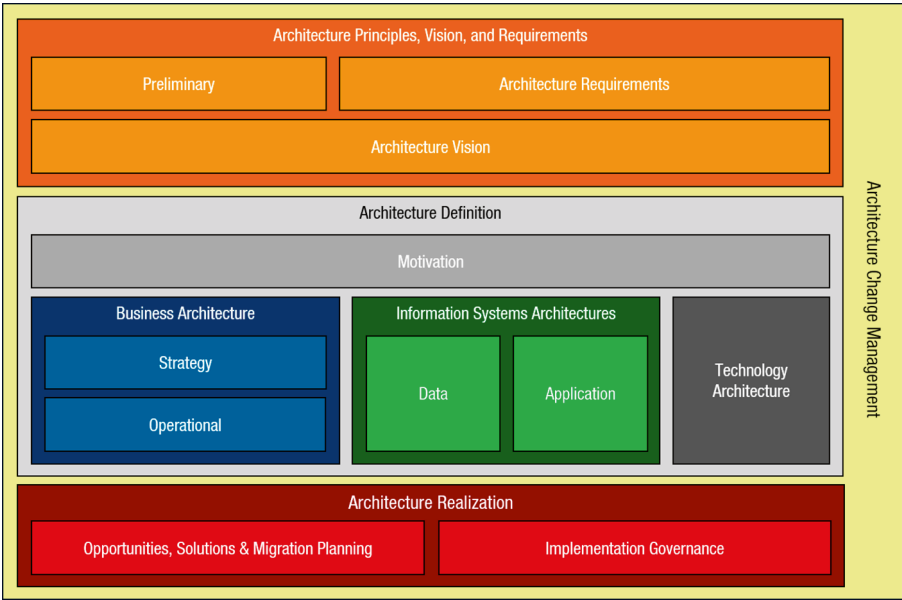


Figure 6-1. *The Content Framework by ADM phase*

The Content Framework consists of the following sections:

Architecture Principles, Vision, Motivation, and Requirements:

The models in this section are intended to capture the surrounding context of formal architecture models, including general Architecture Principles, strategic context that forms input for architecture modeling, and requirements generated from the architecture.

The relevant aspects of the business context that have given rise to the Request for Architecture Work are typically recorded in the Preliminary and Architecture Vision phases.

Business Architecture: This part of the Content Framework captures architecture models of the business, looking specifically at factors that motivate the organization, its structure, and its capabilities. Phase B (Business Architecture) of the Architecture Development Method aligns with this part of the Content Framework.

Information Systems Architectures: The models described in this part of the Content Framework capture architecture models of applications and data. The Information Systems Architectures phase is used here.

Technology Architecture: The Content Framework describes models that capture technology assets, which are used to implement and realize information system solutions. This is realized by applying Phase D (Technology Architecture) of the Architecture Development Method.

Architecture Realization: The models in this section capture change roadmaps that show the transition between architecture states. They are used to guide and drive the implementation of the architecture. Phase F (Migration Planning) and Phase G (Implementation Governance) align with this part of the Content Framework.

Architecture Change Management: Internal and external events that affect the Enterprise Architecture are captured in this section of the Content Framework. They generate action requirements. Architecture Change Management is provided by Phase H of the Architecture Development Method.

The Content Framework should be used to *complement* the Architecture Development Method, as it describes what the architecture should look like when it is complete.

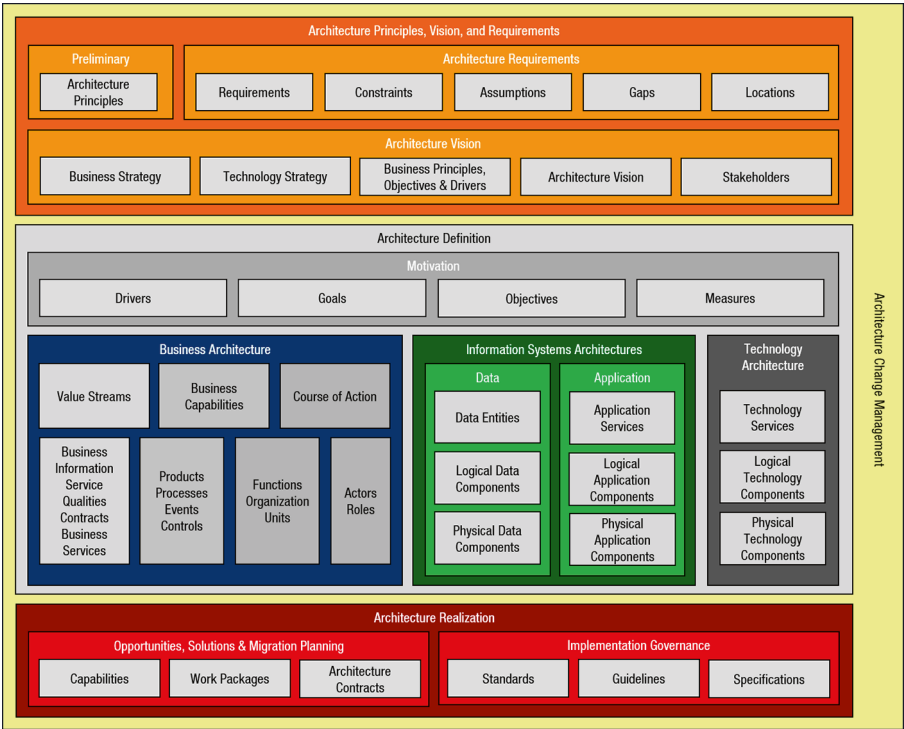


Figure 6-2. *The Extended Content Framework*

The *Extended Content Framework*, as the model in Figure 6-2 is called, contains the entities to be used or applied, in addition to the relationship to the phases of the Architecture Development Method mentioned earlier. The entities used and mapped in the Extended Content Framework are derived from the Enterprise Metamodel, which is described in detail in Chapter 5.

For example, consider the top-left section called Preliminary, which is part of the top layer called Architecture Principles, Vision, and Requirements. The Extended Content Framework shows that the development of Architecture Principles is what unfolds during this phase of the Architecture Development Method. Frankly, there is much more that happens during the Preliminary Phase, but the Extended Content Framework focuses on the development of Architecture Principles.

To the right of the Preliminary section is the Architecture Requirements section. This area contains the entities *requirements*, *constraints*, *assumptions*, *gaps*, and *locations* – most of the General Entities.

Just below the Architecture Principles, Vision, and Requirements section is the Architecture Definition. This section contains by far the most entities. It is divided into the sections Motivation, Business Architecture, Information Systems Architectures, and Technology Architecture. These last three areas naturally relate to phases B, C, and D of the Architecture Development Method.

The section called Motivation differs slightly from the previous one-to-one references to the Architecture Development Method phases. Motivation contains the *driver*, *goal*, *objective*, and *measure* entities, all of which belong to the Business Architecture section according to the Enterprise Metamodel. One would expect these entities to have been made part of the Business Architecture section in the Extended Content Framework, but for some unknown reason the TOGAF Standard has not chosen to do so.

This is one of the inconsistencies in the TOGAF Standard.

Originally, the Business Architecture section had a relatively well-defined set of entities.

However, it may well be that the desire to emphasize the coherence of architectural entities, elements, and concepts in the development of the TOGAF Standard led to the addition of the driver, goal, objective, and measure entities. Add to this the fact that many different people have worked on the Standard, and it is less surprising that such inconsistencies sometimes creep in unintentionally.

The Business Architecture section consists of 19 entities, all of which are described and explained in detail in Section 5.1.2 of Chapter 5. The same goes for the Information Systems Architectures (Data and Application Architecture) and Technology Architecture sections. The entities found in the Data, Application, and Technology Architectures are covered in Chapter 5, Sections 5.1.3, 5.1.4, and 5.1.5, respectively.

The final section of the Extended Content Framework is called Architecture Realization and consists of two parts, Opportunities, Solutions, and Migration Planning on the one hand and Implementation Governance on the other. The Opportunities, Solutions, and Migration Planning section has a one-to-one relationship with phases E and F of the Architecture Development Method. The entities *capabilities* and *work packages* mentioned in this section are part of the General Entities in the Enterprise Metamodel, as are the previously mentioned requirements, constraints, assumptions, gaps, and locations. The name of the Implementation Governance section indicates that it is associated with the phase of the Architecture Development Method of the same name. As an exception to the rule, the entities it contains are not found in the Enterprise Metamodel. The entities *standards*, *guidelines*, and *specifications* are taken from the Architecture Repository (see Chapter 7).

Architecture Change Management, as a section related to Phase H of the Architecture Development Method, is located in the Extended Content Framework on the right-hand side and runs from top to bottom.

The Content Framework and the Extended Content Framework are directly related to the phases of the Architecture Development Method. The TOGAF Standard does this very cleverly by directly showing that the schematic representation of an architectural approach can be seamlessly related to the well-known approach for its implementation, the Architecture Development Method.

6.4. Tailoring the Content Framework

As mentioned in Chapter 5, Section 5.2, the Architecture Development Method can update the Content Framework and the Enterprise Metamodel, but only under specific circumstances. While the Content Framework is a set of reference materials, the iterative nature of the Architecture Development Method allows the architecture content to be updated and refined as the architecture project progresses, specifically during the following phases:

Phase A (Architecture Vision): The initial architecture content is established and aligned with the business objectives. This phase may identify the need for additional content or specific viewpoints that weren't originally part of the Content Framework.

Phases B, C, and D (Business, Information Systems, and Technology Architectures): As detailed architectures are developed in these phases, the specific architecture artifacts created may add to or modify existing components of the Content Framework.

Phase H (Architecture Change Management): This phase is explicitly designed to manage changes in the architecture, which may necessitate updates to the Content Framework to reflect new or modified architecture standards, models, or deliverables.

Throughout the Architecture Development Cycle: New insights, stakeholder feedback, or the identification of new architectural requirements may drive changes or additions to the Content Framework, but to the Enterprise Metamodel as well.

For example, during Phase F of the Architecture Development Method, it is recommended that a roadmap be created that includes all the activities required to implement the architecture work. Activities in the context of the TOGAF Standard are referred to as *initiatives* or *work packages*.

The Content Framework and the Enterprise Metamodel both include a reference to the work package entity. However, the TOGAF Standard doesn't seem to use this entity in any way other than in the Project Context Diagram artifact (see Chapter 12, Section 12.8.2, Figure 12-41).

It is therefore recommended that an additional architecture artifact be included in the Content Framework. Specifically, this would be a Work Package Portfolio Map (see Chapter 14, Section 14.8). It would also be a good idea to include the Work Package Portfolio Map artifact in the model as shown in Figure 7-13 in Chapter 7, Section 7.5.1.

During Phase E (Opportunities and Solutions) and Phase F (Migration Planning) of the Architecture Development Method, the Work Package Portfolio Map comes in handy. This artifact contains all the initiatives (work packages) needed to perform the architecture work. Additionally, this artifact illustrates the relationships between initiatives and organizational goals and objectives. Therefore, the Work Package Portfolio Map is an essential artifact for establishing the relationship between the organization's business goals and their planning using the Architecture Roadmap.

While the Enterprise Metamodel and the Content Framework serve as a guide, the iterative execution of the Architecture Development Method ensures that they evolve and are refined as part of the overall architecture governance and development process.



Further Reading

For more information on customizing the Content Framework, see “The TOGAF® Leader’s Guide to Establishing and Evolving an EA Capability”, Chapter 8, Customization of Architecture Contents and Metamodel, document number G184.

6.5. Summary

This chapter described the Content Framework, an important part of the TOGAF Standard. The Content Framework is a categorization framework that is used to make sense of descriptions of architectures. It is also used to place and clarify the different types of entities in the context of the Architecture Development Method. The Content Framework is a one-to-one translation of the phases of the Architecture Development Method.

In addition to the compact form of the framework, there is also a more extensive form called the Extended Content Framework. In this variant, all entities used in the architecture work are mapped to the phases of the Architecture Development Method. The entities used and mapped in the Extended Content Framework are derived from the Enterprise Metamodel.

The Content Framework aims to be more adaptable to various organizational contexts while ensuring that all relevant architecture elements are covered comprehensively. It is more modular than previous versions, making it more adaptable. It also defines key artifacts like catalogs, matrices, and diagrams that help document different aspects of the architecture. The framework emphasizes the creation of views and viewpoints and places a stronger emphasis on supporting agile methodologies and digital transformation efforts.

A major shift in the TOGAF Standard when compared with earlier versions is its focus on outcomes rather than just processes. The Content Framework reflects this by emphasizing the creation of valuable architecture artifacts that deliver business outcomes, rather than just following a rigid structure.

The Content Framework differs from previous versions in *flexibility*, *integration*, and its emphasis on *outcomes*. The modular and adaptable nature of the TOGAF Standard makes the Content Framework more flexible than in previous versions. It better integrates with other frameworks and methodologies such as the agile methodology. There is a stronger focus on creating actionable outputs that directly contribute to achieving business objectives.